VANISHING OF H^1 OF THE STRUCTURE SHEAF ON ONE CURVE FIBRE OF A PROPER MAP SPREADS TO A NEIGHBOURHOOD, UNIVERSALLY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 20942-20949 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 20951-20968, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. This is the input lemma cited inside the collected card L16073 (globally generated vanishing); registered here as its own proof.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Upper semicontinuity input for fibre dimension (Morphisms, lemma-openness-bounded-dimension-fibres) and the proper cohomology finiteness plus base-change theorems (Limits, lemmas proper-top-cohomology-finite-type, higher-direct-images-zero-above-dimension-fibre).

Source reference. Original lemma 0E7D: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20968–20968. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0EX4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20968–20968. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02FZ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20968–20968. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a proper morphism of schemes. Let $y \in Y$ be a point with $\dim(X_y) \leq 1$ and $H^1(X_y, \mathcal{O}_{X_y}) = 0$. Then there is an open neighbourhood $V \subset Y$ of y such that $R^1 f_* \mathcal{O}_X|_V = 0$ and the same is true after base change by any $Y' \to V$.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. To prove the lemma we may replace Y by an open neighbourhood of y. Thus we may assume Y is affine and that all fibres of f have dimension ≤ 1, see Morphisms, Lemma 02FZ. In this case $R^1 f_* \mathcal{O}_X$ is a quasi-coherent $\mathcal{O}_Y$-module of finite type and its formation commutes with arbitrary base change, see Limits, Lemmas 0EX4 and 0E7D. Say $Y = \mathrm{Spec}(A)$, y corresponds to the prime $\mathfrak{p} \subset A$, and $R^1 f_* \mathcal{O}_X$ corresponds to the finite A-module M. Then $H^1(X_y, \mathcal{O}_{X_y}) = 0$ means that $\mathfrak{p} M_{\mathfrak{p}} = M_{\mathfrak{p}}$ by the statement on base change. By Nakayama's lemma we conclude $M_{\mathfrak{p}} = 0$. Since M is finite, we find an $f \in A$, $f \notin \mathfrak{p}$ such that $M_f = 0$. Thus taking V the principal open $D(f)$ we obtain the desired result. $\square$